AgroEdge AI: An Edge AI-Based Foliar Pathology and Autonomous Precision Agronomy Monitoring System

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***Abstract*—AgroEdge AI is an edge-native artificial intelligence web application that diagnoses foliar crop diseases and automates closed-loop precision irrigation using on-device neural inference and real-time environmental telemetry. Standard cloud-dependent smart agriculture architectures suffer from high latency, recurring cellular bandwidth expenses, and complete system failure under erratic rural connectivity. AgroEdge AI eliminates cloud dependency during inference by deploying a lightweight MobileNet architecture fine-tuned on the 38-class PlantVillage benchmark dataset (54,303 leaf images). Four convolutional architectures were evaluated, and the INT8 Post-Training Quantized MobileNet model achieved the lowest recorded memory footprint of 2.56 MB (a 75% reduction from 14.0 MB) while accelerating CPU inference latency to 18 ms. The model attained a recorded top-1 diagnostic accuracy of 96.4%. To deliver actionable utility, a deterministic agronomic remediation engine prescribes dual-pathway (bio-organic and chemical) treatments with dosage metrics, while an autonomous closed-loop threshold module actuates solenoid irrigation valves based on volumetric soil moisture levels (<30%). The application is implemented using Python, ONNX Runtime, and Streamlit, and is deployed live on Streamlit Community Cloud. AgroEdge AI demonstrates an end-to-end engineering workflow involving dataset preprocessing, transfer learning, quantization, edge acceleration, autonomous actuation, and production deployment. Predictions are automated decision-support estimates and should be accompanied by standard agronomic scouting.**

***Keywords*—Plant disease detection, Edge AI, MobileNet, Post-Training Quantization, ONNX Runtime, Closed-Loop Irrigation, Streamlit, Precision Agriculture.**

I. Introduction

A. Background

Agricultural crop yield is vulnerable to severe destruction caused by foliar fungal, bacterial, and viral pathogens. According to the Food and Agriculture Organization (FAO), plant diseases cost the global economy over $220 billion annually, causing between 20% and 40% of agricultural production loss. Traditional manual scouting by agronomists is labor-intensive, infrequent, and subjective, often leading to delayed treatment and extensive crop necrosis. While recent advances in deep learning have enabled automated image classification of plant diseases, the vast majority of existing solutions rely exclusively on centralized cloud computing servers.

In practical agricultural environments, this cloud-dependent model encounters critical barriers: rural farmlands, greenhouses, and orchards routinely suffer from absent or unreliable internet connectivity. Furthermore, autonomous field systems—such as robotic sprayers and aerial drones—require millisecond-level decision making that cannot tolerate round-trip network delays. AgroEdge AI resolves these bottlenecks by shifting deep learning inference directly to the edge, executing quantized models locally on CPU runtimes with zero cloud dependence.

B. Project Overview

AgroEdge AI is a full-stack, edge-enabled precision farming and foliar pathology platform. It accepts foliar leaf photography via local file uploads, preset sample libraries, or real-time camera video streams, preprocessing the imagery into a standardized 224x224 RGB tensor. An INT8 quantized MobileNet model running on ONNX Runtime classifies the image across 38 distinct crop disease categories in 18 ms. The system simultaneously executes an agronomic remediation database providing organic and chemical prescriptions, a closed-loop soil moisture actuation engine controlling virtual solenoid valves, and an IoT sensor telemetry buffer.

C. Problem Statement

Smallholder farmers and precision growers lack accessible, rapid, and offline-capable diagnostic tools to identify foliar pathologies before they spread. Existing smart farming solutions enforce continuous broadband internet connectivity, incur expensive data fees, and introduce unacceptable latency. There is an acute need for an edge-native, lightweight AI platform that performs sub-30ms offline foliar diagnosis, prescribes immediate scientific remedies, and automates soil hydration without requiring cloud infrastructure.

D. Objectives

- To develop an edge-native foliar pathology classifier operating locally on CPU/ARM hardware with zero cloud dependency.
- To evaluate transfer learning across deep convolutional architectures and apply INT8 Post-Training Quantization (PTQ) to minimize parameter memory footprint below 3.0 MB.
- To achieve sub-30ms CPU inference latency while maintaining greater than 95% classification accuracy across 38 agricultural plant disease classes.
- To engineer an agronomic remediation knowledge base prescribing dual-pathway (bio-organic and chemical fungicide) interventions with dosage guidelines.
- To construct a closed-loop soil moisture and microclimate decision engine that automates solenoid irrigation valve actuation and spore risk alerts.
- To deploy the integrated platform as an interactive, production-grade web application with automated CI/CD testing on GitHub Actions and public cloud hosting.

II. Need and Societal Impact

A. Need for the Project

A practical agricultural AI platform must deliver more than raw statistical classifications. Farmers and agronomists need immediate, verifiable treatment protocols to arrest disease progression and protect seasonal investments. AgroEdge AI merges edge vision classification, evidence-based chemical/biological prescriptions, and automated irrigation management into an intuitive, responsive interface.

B. Societal Impact

By eliminating reliance on high-speed cellular infrastructure, AgroEdge AI democratizes access to state-of-the-art agronomy for smallholders in developing regions. Precision chemical dosage guidance prevents excessive pesticide runoff into local water tables, and closed-loop soil irrigation saves up to 40% of agricultural fresh water.

C. Alignment with Sustainable Development Goals

- **SDG 2: Zero Hunger:** Mitigating crop losses through early foliar disease intervention safeguards food supply chains.
- **SDG 6: Clean Water and Sanitation:** Closed-loop soil moisture actuation eliminates excessive irrigation and prevents chemical fertilizer leaching.
- **SDG 12: Responsible Consumption and Production:** Precise fungicide recommendations reduce over-application of synthetic pesticides.

III. Dataset Description

A. Sources of Data

The computer vision model was trained and evaluated on the internationally recognized **PlantVillage** benchmark dataset (Mohanty et al.), coupled with agronomic soil telemetry parameters (Crop and Fertilizer Recommendation records).

TABLE I
Data Sources

No.	Dataset / Asset Name	Format & Scale	Purpose
1	PlantVillage Vision Dataset	54,303 Images (38 Classes)	Historical foliar records for model development
2	Crop & Soil Telemetry Dataset	Tabular (NPK, Moisture, Temp, pH)	Irrigation threshold and soil health modeling
3	Agronomic Pathology Database	38 Curated Scientific Profiles	Organic remedies, fungicides, and prevention guidelines
4	Quantized ONNX Model	2.56 MB Binary (.onnx)	Consistent edge execution of new input records

B. Dataset Size and Features

The computer vision dataset comprises 54,303 annotated leaf images across 14 agricultural crops: Apple, Blueberry, Cherry, Corn (Maize), Grape, Orange, Peach, Bell Pepper, Potato, Raspberry, Soybean, Squash, Strawberry, and Tomato. The dataset covers 26 disease categories (bacterial spots, early/late blights, rusts, powdery mildews, leaf scabs, viral curls) and 12 healthy foliar classes, forming a comprehensive 38-class taxonomic challenge.

IV. Dataset Preprocessing & Edge Optimization

Preprocessing prepares submitted leaf imagery and environmental parameters into the exact numerical format expected by the model. The project encapsulates these transformations into modular Python pipelines loaded during application startup.

- **Geometric Standardization:** All images are coerced to 3-channel RGB and resized to 224x224 pixels using bilinear interpolation.
- **Channel Normalization:** Raw 8-bit integer pixel intensities [0, 255] are rescaled to [0.0, 1.0] and standardized against ImageNet mean ([0.485, 0.456, 0.406]) and standard deviation ([0.229, 0.224, 0.225]).
- **Tensor Layout Permutation:** The tensor memory layout is converted from standard (H, W, C) to channel-first (1, 3, 224, 224) format for optimal memory alignment in ONNX Runtime linear algebra kernels.
- **Post-Training Quantization (INT8):** 32-bit floating-point weights (FP32) are transformed into 8-bit signed integers (INT8) using symmetric uniform quantization. This reduces the binary footprint from 14.0 MB to 2.56 MB (a 75% compression ratio) with negligible accuracy degradation (<0.8%).

V. Models and Technologies Used

The project evaluated four supervised deep learning architectures: VGG-16, ResNet-50, MobileNetV2 (FP32), and MobileNetV2 (INT8 Quantized). VGG-16 establishes an architectural baseline, ResNet-50 demonstrates residual learning, MobileNetV2 introduces depthwise separable convolutions, and MobileNetV2 INT8 applies 8-bit integer quantization.

TABLE II
Model Benchmark Comparison

Model Architecture	Precision	File Size	CPU Latency	Top-1 Accuracy
VGG-16 (Baseline)	FP32	138.4 MB	284 ms	96.8%
ResNet-50	FP32	97.5 MB	142 ms	97.4%
MobileNetV2 (Standard)	FP32	14.0 MB	65 ms	97.1%
AgroEdge AI MobileNet (Ours)	INT8 Quantized	2.56 MB (-75%)	18 ms (3.6x faster)	96.4% (-0.7%)

A. Selected Deep Learning Model

MobileNetV2 INT8 Quantized was selected because it achieved the lowest recorded latency (18 ms) and smallest binary footprint (2.56 MB) while maintaining an exceptional 96.4% top-1 accuracy. While ResNet-50 achieved a marginally higher accuracy of 97.4%, its 97.5 MB footprint and 142 ms latency make it unviable for energy-constrained edge deployment. The selection is based primarily on edge efficiency and speed.

B. Technologies Used

TABLE III
Technologies Used

Component	Technology / Library	Purpose
Programming	Python 3.11+	Core application and machine learning logic
Edge AI Runtime	ONNX Runtime v1.31	High-performance quantized INT8 model inference
Image Processing	Pillow (PIL), NumPy	Image decoding, resizing, and array transformation
Web Application	Streamlit v1.56	Interactive user interface and live camera capture
Data Analytics	Pandas, Altair	Diurnal telemetry graphs and event logging
Version Control & CI	Git, GitHub Actions	Automated testing and continuous integration
Deployment	Streamlit Community Cloud	Hosting the live public web application

VI. Proposed Methodology and System Architecture

A. Proposed Methodology

The system follows an edge-native supervised classification workflow. Historical plant leaf records are used to fine-tune the backbone architecture. The trained model is converted to ONNX and quantized into INT8 precision. During deployment, submitted leaf images are preprocessed and classified by the local ONNX engine. Class probabilities are mapped to specific agronomic treatments, while a separate rule-based threshold component regulates irrigation valve states based on sensor telemetry.

B. System Architecture

The architecture consists of a client interface, preprocessing module, quantized ONNX inference engine, agronomic prescription repository, and closed-loop irrigation controller. The frontend collects imagery and sensor readings. The inference engine computes local predictions, and results are presented through the interactive dashboard.

C. Working of the System

1. The user captures a crop leaf photo via live camera or uploads an image file.
2. Streamlit passes the image to the local preprocessor, which resizes and normalizes the tensor to (1, 3, 224, 224).
3. The quantized ONNX MobileNet engine executes feed-forward inference on the CPU in 18 ms.
4. Numerically stable Softmax extracts class probabilities, identifying the disease category and confidence percentage.
5. The prescription engine retrieves tailored organic and chemical interventions with dosage safety guidelines.
6. The irrigation advisory module evaluates soil moisture levels (<30%) and actuates virtual solenoid valves automatically.
7. The results dashboard renders diagnostic assessments, confidence progress bars, treatment plans, and telemetry curves.

VII. Implementation

A. Frontend and Backend

The frontend and application controller are implemented using Streamlit with custom CSS. The core interface is defined in **app.py**, which encapsulates navigation, KPI metric cards, specimen drawers, and telemetry graphs. The modular edge engine resides in **src/classifier.py**, remedies in **src/remedies.py**, and irrigation logic in **src/soil_advisor.py**.

B. Edge AI Inference Engine

The application utilizes *onnxruntime.InferenceSession* with CPUExecutionProvider. Thread count is locked to 4 threads for deterministic embedded execution. Preprocessing normalizes the input using standard ImageNet parameters, and numerically stable Softmax prevents numerical overflow.

C. Agronomic Prescription Engine

Recommendations are generated by a domain-specific agronomic knowledge base covering all 38 classes. It details target fungal/bacterial pathogens, organic bio-fungicides (e.g., neem oil, Bacillus subtilis), chemical fungicides (e.g., Mancozeb, Chlorothalonil), and field hygiene.

D. Closed-Loop Irrigation Actuator

Automated irrigation logic operates via a closed-loop threshold controller. Volumetric soil moisture below 30% triggers the virtual solenoid valve to OPEN (pumping water at 45 L/min). Moisture between 30% and 60% maintains STANDBY mode, conserving water.

E. Deployment

The application is deployed on Streamlit Community Cloud at **https://agroedgeai.streamlit.app**. Source code and CI/CD pipelines are maintained at **https://github.com/rishu295/AgroEdge-AI**. The deployment requires only the lightweight repository and dependencies listed in requirements.txt.

VIII. Results and Evaluation

A. Evaluation Metrics

Models were evaluated using Top-1 Accuracy, Inference Latency (ms), Storage Size (MB), and Compression Ratio. Accuracy measures correct pathology identifications on test data, while latency and storage determine real-world edge viability.

TABLE IV
AgroEdge AI Performance Results

Evaluation Metric	Target Requirement	AgroEdge AI Recorded Result	Verdict
Top-1 Classification Accuracy	> 95.0%	96.4% on PlantVillage	Met (High Reliability)
CPU Inference Latency	< 30 ms	18.2 ms (Intel/AMD x86)	Met (Real-Time Capable)
Model Memory Footprint	< 5.0 MB	2.56 MB (INT8 ONNX)	Met (75% Compression)
Offline Resilience	100% Offline Inference	0 Cloud API Calls Required	Met (Full Edge Autonomy)
Irrigation Actuation Speed	< 100 ms	< 2 ms (Local Rule Engine)	Met (Instant Closed-Loop)

B. Results Discussion

AgroEdge AI achieved a 96.4% top-1 accuracy across 38 classes while reducing latency from 65 ms to 18 ms and shrinking model size from 14.0 MB to 2.56 MB. The 0.7% accuracy trade-off is minor compared to the substantial 3.6x speedup and 75% memory reduction achieved.

IX. Comparison with Other Approaches

TABLE V
Comparison of Approaches

Approach	Strengths	Limitations

Manual Scouting	Direct expert physical inspection	Slow, labor-intensive, subjective, costly
Cloud-Based AI APIs	Massive server-side deep learning models	Fails without internet; recurring cellular data fees
Satellite Remote Sensing	Broad macro-level regional monitoring	Low foliar resolution; cannot detect micro-lesions
Uncompressed Edge CNNs	High raw classification accuracy	Large file sizes (>100MB), drains battery, high heat
AgroEdge AI (Ours)	Sub-20ms offline inference, 2.56MB size	Requires clear top-down leaf imagery framing

X. Individual Contribution

The project work covered problem definition, PlantVillage dataset curation, MobileNet transfer learning, INT8 Post-Training Quantization, ONNX Runtime optimization, agronomic prescription engine design, closed-loop soil moisture logic, Streamlit web development, GitHub Actions CI/CD configuration, deployment on Streamlit Community Cloud, and academic report compilation.

XI. Limitations

• Laboratory background bias: PlantVillage images were captured under controlled conditions, which may slightly reduce accuracy under complex field backgrounds.

• Single-leaf constraint: The classification model evaluates individual leaves rather than full-canopy aerial imagery.

• Lack of calibrated confidence intervals: Predictions provide softmax point estimates rather than Bayesian uncertainty distributions.

• Soil moisture threshold generalization: Standard 30%–60% field capacity thresholds may require regional tuning for sandy or heavy clay soils.

• Heuristic recommendations: Agronomic prescriptions are rule-based and have not been dynamically adjusted for newly evolving resistant pathogen strains.

XII. Future Scope

• Integrate YOLOv11-Nano object detection for multi-leaf bounding box detection and lesion segmentation in live video feeds.

• Deploy the INT8 ONNX model onto physical Raspberry Pi 5 and ESP32-CAM hardware with physical 12V solenoid relay valves.

• Integrate LoRaWAN mesh networking for long-range (10 km) sensor telemetry communication without cellular infrastructure.

• Develop multilingual voice advisory features for non-literate smallholder farmers in regional languages.

XIII. Conclusion

AgroEdge AI demonstrates an end-to-end implementation of Edge AI for automated foliar pathology and smart precision irrigation. Using a fine-tuned MobileNet architecture compressed via INT8 Post-Training Quantization into an ONNX graph, the system achieves a 2.56 MB footprint (75% compression) and an ultra-fast 18 ms inference latency while retaining 96.4% diagnostic accuracy across 38 classes. Coupled with evidence-based agronomic treatment guidelines, an autonomous closed-loop irrigation actuator, and a responsive web application deployed online, AgroEdge AI proves that modern deep learning can operate reliably and offline to support global agricultural resilience.

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